

Hybrid Adaptive Kalman Filtering for Data-Efficient Joint Tracking and Classification

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Abstract—Kalman filtering performance is highly sensitive to model mismatch and noise covariance tuning. Classical adaptive filtering methods address these issues by estimating model parameters or noise covariances from measurement data. However, these methods often assume a correctly specified model structure, and structural mismatch that falls outside the assumed model induces bias and temporally correlated innovations that parameter adaptation alone may not fully correct. Learning-based filtering methods provide greater flexibility but typically rely on supervised training with large datasets and do not produce consistent uncertainty estimates. This article proposes a self-supervised Hybrid Adaptive Kalman Filter that integrates classical adaptive filtering principles with data-driven methods, learning structured corrections to system dynamics and process noise covariance from measurements while preserving the probabilistic structure of the filter. This allows the innovation likelihood to be computed and subsequently used for model classification via generalized Bayesian inference. Experimental results on real-world and high-fidelity simulated drone tracking datasets demonstrate improved estimation accuracy and statistical

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consistency as well as robust classification performance across both low-data and large-data scenarios.

Nomenclature

Symbol	Definition
x_k	State vector at time step k .
z_k	Measurement vector at time step k .
F_k	State transition matrix.
F_k^{MB}	Nominal model-based state transition matrix.
ΔF_k^{ML}	Learned correction to the state transition matrix.
H_k	Observation matrix.
Q_k	Process noise covariance matrix.
Q_k^{ML}	Learned process noise covariance matrix.
R_k	Measurement noise covariance matrix.
w_k	Process noise, $w_k \sim \mathcal{N}(0, Q_k)$.
v_k	Measurement noise, $v_k \sim \mathcal{N}(0, R_k)$.
$\hat{x}_{k k-1}$	Predicted state estimate.
$P_{k k-1}$	Predicted state covariance.
$\hat{x}_{k k}$	Posterior state estimate.
$P_{k k}$	Posterior state covariance.
ν_k	Innovation (measurement residual).
S_k	Innovation covariance.
K_k	Kalman gain.
$e_{x,k}$	State estimation error.
$\Delta \hat{x}_k$	State correction from measurement update.
$\nu_k^{(K)}$	Window of recent innovations.
ϕ_k	Feature vector used by the adaptive neural network.
$f_\theta(\cdot)$	Neural network parameterized by θ .
λ	Interpolation parameter between nominal and learned dynamics.
$L(\theta)$	Training loss function.
M_j	Candidate model or class j .
α_j	Tempering parameter for generalized Bayesian inference.
$\epsilon_{x,k}$	Normalized estimation error squared.
$\epsilon_{z,k}$	Normalized innovation squared.

I. Introduction

Kalman filters (KFs) and their variants have been widely used for tracking, navigation, and state estimation because of their principled probabilistic framework and computational efficiency [1], [2], [3]. In practice, however, the performance of a KF depends heavily on accurate system models and carefully tuned noise covariances. Model mismatch, unmodeled disturbances, and mistuned noise statistics can lead to biased state estimates, inconsistent uncertainty quantification, and degraded downstream inference. Obtaining reliable model parameters often requires expert knowledge, system identification, and manual tuning.

Classical adaptive filtering methods, extensively studied since the 1970s, address these challenges by adapting filter parameters directly from measurement data

[4]. Techniques such as covariance matching, correlation analysis, and likelihood-based estimation enable parameter adaptation without requiring ground-truth state information [5], providing an important foundation for measurement-based filter tuning. However, they typically assume a predefined parametric state-space structure. As a result, structural mismatch that falls outside the assumed model can induce biases and temporally correlated innovations that cannot be corrected through parameter adaptation alone.

Recent advances in machine learning have motivated a growing body of work that integrates neural networks with classical filtering frameworks [6]. These learning-augmented approaches aim to improve estimation performance by learning system dynamics, filter parameters, or correction terms directly from data. While promising, existing learning-based filtering methods typically rely on supervised training with access to ground-truth states and are optimized using accuracy-based loss functions. As a result, these methods primarily improve point-estimation accuracy but often neglect the statistical consistency of the estimated uncertainty, which is critical for assessing filter reliability. Furthermore, many learning-based approaches assume access to large-scale training datasets, often generated in simplified simulation environments, which may not be available in real-world systems.

These limitations pose challenges for many safety-critical systems where ground-truth state information may be unavailable, uncertainty calibration is essential, and collecting large labeled datasets can be costly or infeasible. Motivated by these limitations, this article proposes a Hybrid Adaptive Kalman Filter (HAKF) that combines model-based filtering with data-driven learning. The proposed framework augments a classical KF with a neural network that learns structured corrections to the nominal dynamics and noise covariances directly from measurement data. By incorporating structural inductive bias from the underlying physics-based model and restricting learning to components most susceptible to mismatch, the proposed approach enables data-efficient adaptation while preserving the probabilistic structure of the filter.

In addition to improving state estimation, the measurement likelihood produced by the hybrid filter naturally enables model classification through a generalized Bayesian inference (GBI) framework [7]. This allows the same filtering architecture to perform both probabilistic state estimation and model classification. Fig. 1 illustrates the overall conceptual differences between classical KF, purely learning-based approaches, and the proposed hybrid framework.

We evaluate the proposed hybrid method on both real-world drone trajectories from the multimodal indoor 3D drone tracking dataset (DPJAIT) [8] and high-fidelity simulated trajectories generated using AirSim [9]. Experimental results demonstrate that HAKF improves estimation accuracy while achieving substantially better statistical consistency compared to a tuned classical KF. For model classification, HAKF shows strong perfor-

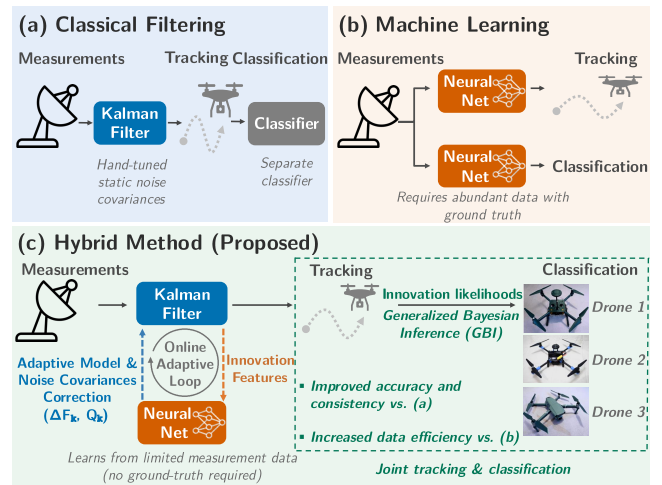


Fig. 1. Comparison between classical filtering, purely learning-based methods, and the proposed hybrid framework. (a) Classical KF relies on manually tuned noise covariances and domain expertise to produce reliable state estimates and uncertainty. (b) Purely learning-based approaches use neural networks to directly predict system states, typically requiring abundant ground-truth training data. (c) The proposed Hybrid Adaptive Kalman Filter combines model-based filtering with data-driven adaptation, enabling accurate state estimation and probabilistic model classification under limited-data conditions.

mance in low-data regimes where purely learning-based methods struggle, while also remaining effective when larger training datasets are available.

To summarize, the key contributions of this article are as follows.

- 1) A self-supervised HAKF framework inspired by classical adaptive KF that integrates model-based filtering with neural network-based adaptive model correction without requiring ground-truth state supervision.
- 2) A GBI framework that enables probabilistic model classification using the innovation likelihood produced by the hybrid filter.
- 3) Experimental validation on both real-world and simulated datasets demonstrating robust performance spanning the low-data and large-data regimes.

The rest of this article is organized as follows. Section II reviews related work, and Section III presents the proposed HAKF framework. Section IV describes the experimental setup, including the system model, datasets, evaluation metrics, and baseline methods, and Section V reports experimental results for both state estimation and model classification. Finally, Section VI concludes the articles and discusses limitations and directions for future work.

II. Related Work

A. Kalman Filter (KF)

The KF provides a Bayesian optimal estimator for linear dynamical systems subject to Gaussian noise [1].

Consider the following discrete-time state-space model:

$$x_k = F_k x_{k-1} + w_k, \quad (1)$$

$$z_k = H_k x_k + v_k. \quad (2)$$

The Nomenclature summarizes the KF notation used throughout the article.

If the linear-Gaussian assumptions of the KF are satisfied and the system model is correctly specified, then the following dynamical consistency conditions must hold [2]:

1) Unbiased

$$\mathbb{E}[e_{x,k}] = 0 \quad \forall k. \quad (3)$$

2) Efficiency

$$\mathbb{E}[e_{x,k} e_{x,k}^\top] = P_{k|k} \quad \forall k. \quad (4)$$

3) Innovation Whiteness

$$\nu_k \sim \mathcal{N}(0, S_k), \quad (5)$$

$$\mathbb{E}[\nu_k] = 0, \quad (6)$$

$$\mathbb{E}[\nu_k \nu_j^\top] = \delta_{kj} S_k. \quad (7)$$

In practice, the assumed linear-Gaussian system model is often incorrect due to modeling errors and mistuned noise covariances. Such structural mismatch leads to biased estimation errors and temporally correlated innovation sequences [5], violating the dynamical consistency conditions (3)–(7) required for optimal Bayesian inference. Hence, violations of these properties provide informative signals of model mismatch that can be exploited for adaptive model correction.

B. Adaptive Kalman Filter

To address this issue, adaptive Kalman filtering methods have been extensively studied since the 1970s [4]. Existing approaches can be broadly categorized into Bayesian inference, maximum likelihood estimation, covariance-matching, and correlation methods. More recently, consistency-based methods have been proposed that tune filter noise covariances by enforcing their statistical consistency.

Bayesian methods treat unknown model parameters as latent random variables and estimate them jointly with the system state through recursive inference. The posterior distribution of the parameters is updated sequentially using Bayes' rule; however, exact inference is generally intractable and typically requires approximate techniques like Monte Carlo sampling, which can introduce computational overhead and limit real-time application [10], [11], [12].

Maximum likelihood methods estimate unknown model parameters by maximizing the likelihood of the observed measurement sequence under an assumed probabilistic model [13], [14]. In the marginal likelihood formulation, the state variables are treated as latent and integrated out, and parameters are obtained by solving

$$\theta^* = \arg \max_{\theta} p_{\theta}(z_{1:T}), \quad (8)$$

where θ denotes the set of unknown model parameters. These methods typically require iterative optimization of a non-convex likelihood objective, making them sensitive to initialization and susceptible to local optima convergence [5].

Covariance-matching techniques enforce consistency between the empirical covariance of the innovation sequence and its theoretical covariance predicted by the KF. More formally,

$$\frac{1}{m} \sum_{i=k-m+1}^k \nu_i \nu_i^\top \approx S_k, \quad (9)$$

where m is an empirically chosen finite window length used for statistical smoothing [4].

Correlation-based methods exploit the fact that, under correction model specification, the innovation sequence should be temporally uncorrelated. Nonzero innovation autocorrelation provides a direct signal of misspecification, which can be used to adapt filter parameters [4], [5].

Lastly, recent work proposes consistency-based methods that tune the filter noise covariances such that the state estimation errors and innovations are statistically consistent [15], [16]. The normalized estimation error squared (NEES) and normalized innovation squared (NIS) statistics,

$$\epsilon_{x,k} = e_{x,k}^\top P_{k|k}^{-1} e_{x,k}, \quad (10)$$

$$\epsilon_{z,k} = \nu_k^\top S_k^{-1} \nu_k, \quad (11)$$

follow chi-squared distributions under correct model specification [2]. These methods adjust the noise parameters by matching the empirical statistics to their corresponding theoretical moments.

Our proposed HAKF architecture integrates principles from classical adaptive filtering methods, including maximum likelihood, covariance matching, and correlation approaches, to enable statistically consistent adaptation without requiring ground-truth states. Accordingly, NEES and NIS are used in the experiments to evaluate the consistency of the resulting estimates.

C. Hybrid Kalman Filter

Recent work combines KF with learned neural components to mitigate model mismatch and unknown noise statistics. These approaches are commonly referred to as hybrid methods and can be categorized into the following three categories: state-correction, model-learning, and gain-learning methods.

State-correction approaches augment the Kalman filtering framework with a learned neural component that compensates for estimation errors. In these methods, the neural network typically learns a residual mapping to correct either the predicted state or the innovation term without modifying the assumed system model [6], [17], [18]. While such approaches can improve point-estimation accuracy, they rely on post hoc compensation and do not

explicitly enforce statistical consistency of the associated estimates.

Model-learning methods learn unknown system parameters directly from data. In these approaches, neural networks parameterize components of the underlying state-space model, enabling the representation of complex nonlinear dynamics beyond classical linear-Gaussian assumptions [19], [20], [21]. This improves modeling flexibility but often replaces explicitly structured model components with learned parameterizations, making the resulting filter harder to interpret in classical state-space terms.

Lastly, gain-learning methods retain the prediction step of the KF while replacing the analytically derived measurement update step with a learned neural update operator. Representative approaches such as KalmanNet and its variants explicitly replace or augment the analytical Kalman gain computation with recurrent neural networks [22], [23], [24], [25]. These methods have been shown to improve state estimation accuracy under partially known dynamics without requiring explicit identification of the process or measurement noise statistics.

In contrast, our proposed HAKF preserves the analytical Kalman prediction and update recursion and uses learning only to adapt structured corrections to the dynamics and process noise. Unlike hybrid methods trained with supervised state-estimation losses, HAKF is trained directly from measurement data alone. This distinction is important because while KF is often interpreted as an minimum mean-square error (MMSE) estimator, it fundamentally performs recursive Bayesian inference by propagating both the posterior mean and covariance. Consequently, training and evaluating hybrid filters solely in terms of point-estimation accuracy disregards the uncertainty information captured by the filter that is essential for assessing its reliability. Moreover, existing hybrid methods employ expressive neural components that are trained assuming access to abundant training data, which are typically generated from simplified simulation environments.

III. HAKF: Hybrid Adaptive Kalman Filter

In this section, we present a self-supervised HAKF that learns corrections to the system dynamics and process noise while preserving the analytic Bayesian filtering structure. We first describe the problem formulation and the proposed hybrid filtering architecture followed by the training objective, classification method, and implementation details.

A. Problem Formulation

In practice, the discrete-time linear dynamical system described in (1)–(2) is often misspecified due to unmodeled dynamics and inaccurate noise statistics. The objective of the proposed HAKF is therefore to adaptively learn corrections to both the nominal system dynamics

and the process noise covariance from observable features. Specifically, at each time step, a neural network parameterized by θ maps a feature vector ϕ_k to

$$(\Delta F_k^{\text{ML}}, Q_k^{\text{ML}}) = f_\theta(\phi_k), \quad (12)$$

where ϕ_k is constructed from innovations and state-update information described in III-B. Accordingly, the system evolution is modeled as

$$x_k = (F_k^{\text{MB}} + \Delta F_k^{\text{ML}}) x_{k-1} + \tilde{w}_k, \quad \tilde{w}_k \sim \mathcal{N}(0, Q_k^{\text{ML}}). \quad (13)$$

The structure of the correction term ΔF_k^{ML} can be strategically designed to control the degree of model adaptation. More specifically, ΔF_k^{ML} may be constrained to follow a predefined sparsity pattern or low-dimensional parameterization, which allows a trade-off between model expressiveness and data efficiency. This allows the HAKF framework to have varying levels of modeling flexibility while avoiding overparameterization under limited training data. The specific parameterization used in this work is described in Section IV-A.

B. Architecture

Fig. 2 illustrates the proposed HAKF architecture. The filter preserves the standard Kalman prediction and measurement correction recursion. We assume the observation model (i.e. H_k, R_k) is known. The same framework could, in principle, be extended to learn measurement parameters as well, but this is left for future work.

At each time step, a feature vector is constructed from a finite window of recent innovations,

$$\nu_k^{(K)} \triangleq \{\nu_{k-K+1}, \dots, \nu_k\}, \quad (14)$$

which is used as input to a feedforward neural network that outputs ΔF_k^{ML} and Q_k^{ML} . In addition, the state correction

$$\Delta \hat{x}_k = \hat{x}_{k|k} - \hat{x}_{k|k-1} \quad (15)$$

is used as a feature. A multilayer perceptron (MLP) is employed for data-efficient adaptation under limited training data, due to its relatively simple architecture.

In the prediction step, the predicted state covariance is also computed using the corrected transition matrix,

$$P_{k|k-1} = (F_k^{\text{MB}} + \Delta F_k^{\text{ML}}) P_{k-1|k-1} (F_k^{\text{MB}} + \Delta F_k^{\text{ML}})^\top + Q_k^{\text{ML}}. \quad (16)$$

The resulting innovation ν_k from the measurement correction step is appended to the window $\nu_k^{(K)}$, closing the feedback loop. The use of innovation-sequence features in the learning module is motivated by the innovation-based training objective described below in Section III-C.

C. Cost Function

Given the hybrid filtering architecture described above, we consider the marginal maximum likelihood

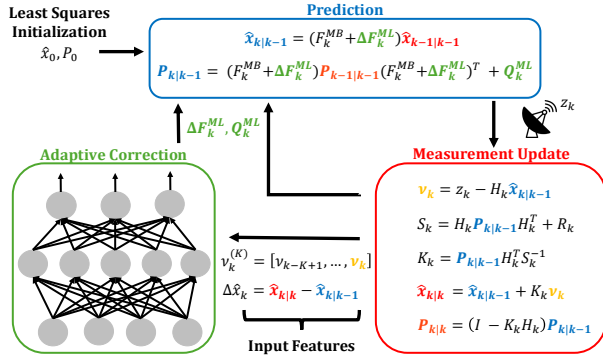


Fig. 2. HAKF architecture. The prediction and measurement correction steps follow the classical KF framework. An adaptive neural network module uses features derived from the innovation sequence and state updates to estimate a transition correction ΔF_k^{ML} and process noise covariance Q_k^{ML} , which are used to compensate for model mismatch in the prediction step.

estimation objective. Under the correct model specification and the KF assumptions, the predictive measurement distribution is

$$z_k | z_{1:k-1} \sim \mathcal{N}(H_k \hat{x}_{k|k-1}, S_k), \quad (17)$$

Substituting this predictive distribution into (8) yields a closed-form expression for the marginal likelihood. Maximizing this likelihood is equivalent to minimizing the innovation negative log-likelihood (NLL), which gives the cost function:

$$\mathcal{L}(\theta) = \sum_{k=1}^T (\nu_k^\top S_k^{-1} \nu_k + \log \det S_k), \quad (18)$$

where an additive constant independent of θ has been omitted.

Under the assumption that the innovations are independent and Gaussian, minimizing the cost function in (18) yields first-order optimality conditions that implicitly enforce covariance-matching on the innovation sequence. This connection has been formally established in the data assimilation literature [26], [27]. Under model misspecification, however, the resulting Kalman gain is generally suboptimal, and the innovation sequence becomes temporally correlated [5], violating the independence assumptions implicit in (18). In this regime, minimizing (18) constitutes quasi-maximum likelihood estimation (QMLE), which converges to the parameters minimizing the Kullback–Leibler divergence between the true innovation process and the assumed independent-Gaussian model [28], [29]. Consequently, the likelihood objective primarily targets the marginal innovation distribution but does not explicitly constrain cross-lag correlations, so temporal dependence can still persist at the likelihood optimum.

To address this limitation, the proposed HAKF architecture uses a finite window of recent innovations as an input feature, enabling the MLP to capture temporal correlation structure that cannot be learned through one recent innovation alone. In this way, HAKF integrates principles from likelihood-based, covariance-matching,

and correlation-based classical adaptive Kalman filtering methods through its cost function and architecture.

D. Classification via Generalized Bayes Inference

The KF can be interpreted as a generative model that induces a predictive measurement likelihood through the distribution defined in (17). Model classification can therefore be mathematically performed by computing the posterior over candidate models using Bayes' rule.

Under model misspecification, however, direct likelihood-based calculation may lead to overconfident or inconsistent posterior estimates [7]. Moreover, in practice, model misspecification may vary across candidate models. To mitigate this issue, we adopt a generalized Bayesian inference (GBI) using a model-dependent tempered likelihood [30],

$$p(M_j | z_{1:k}) \propto p(z_{1:k} | M_j)^{\alpha_j} p(M_j), \quad (19)$$

where $\alpha_j \in (0, 1]$ controls the influence of the likelihood associated with model M_j on the posterior update.

Various approaches for selecting the tempering parameter have been proposed in the literature [31]. In this work, we determine α_j using a grid-based search that minimizes the Brier score on a validation dataset,

$$\alpha_j^* = \arg \min_{\alpha_j} \frac{1}{N} \sum_{i=1}^N \left(p_{\alpha_j}(M_j | \mathcal{Z}^{(i)}) - y_{i,j} \right)^2, \quad (20)$$

where $\mathcal{Z}^{(i)}$ denotes the measurement sequence of the i -th trajectory and $y_{i,j} \in [0, 1]$ indicates whether trajectory i was generated by model M_j . This enables per-class calibration of the predicted model posterior.

E. Implementation Details

For numerical stability and training, several implementation details are adopted. First, the posterior covariance is updated using the Joseph form [32]

$$P_{k|k} = (I - K_k H_k) P_{k|k-1} (I - K_k H_k)^\top + K_k R_k K_k^\top,$$

which guarantees symmetry and positive definiteness of the updated state covariance in the presence of numerical errors.

Second, to ensure that the learned Q_k^{ML} remains positive semi-definite, the neural network outputs a lower-triangular matrix L_k and a diagonal matrix D_k , from which

$$Q_k^{ML} = L_k D_k L_k^\top \quad (21)$$

is constructed via Cholesky LDL composition [33]. In our experiment, this parameterization provided better numerical conditioning during training compared to the commonly used $L_k L_k^\top$ composition in other works [24], [25], [34].

Third, $(\hat{x}_0, P_0)$ is obtained via a weighted least-squares estimate using the minimum number of measurements required to obtain an observable batch estimate of the initial state [3]. Under the assumed motion model

described in Section IV-A, the corresponding solution is given by

$$\hat{x}_0 = (A^\top \Sigma^{-1} A)^{-1} A^\top \Sigma^{-1} Y, \quad (22)$$

$$P_0 = (A^\top \Sigma^{-1} A)^{-1}, \quad (23)$$

where $A = \begin{bmatrix} H \\ H F^{MB} \end{bmatrix}$ and $Y = \begin{bmatrix} y_0 \\ y_1 \end{bmatrix}$ are constructed using two measurements, and Σ denotes a block-diagonal of the stacked measurement noise covariance.

Lastly, the network parameters are optimized using the gradient-based Adam optimizer [35].

IV. Experimental Setup

A. System Model

For experimental evaluation, we consider a constant-velocity (CV) motion model with state

$$x_k = [p_x, p_y, p_z, v_x, v_y, v_z]^\top, \quad (24)$$

where p_x, p_y, p_z denote the 3D position and v_x, v_y, v_z denote the corresponding velocity components. The nominal model-based transition matrix is given by

$$F^{MB} = \begin{bmatrix} I_3 & dt I_3 \\ 0 & I_3 \end{bmatrix}, \quad (25)$$

and the observations are position-only measurements,

$$H = \begin{bmatrix} I_3 & 0 \end{bmatrix}. \quad (26)$$

In the CV motion model, the position update is kinematic and governed by known physics, whereas the constant velocity assumption in the velocity dynamics is often violated in practice due to unmodeled forces and disturbances. Hence, rather than learning a fully data-driven transition model, we restrict learning to a structured correction of the velocity dynamics. Accordingly, the corrected transition matrix is computed as

$$F_k = \begin{bmatrix} I_3 & dt I_3 \\ 0 & (1 - \lambda) I_3 + \lambda \Delta F_k^{ML} \end{bmatrix}, \quad (27)$$

where $\lambda \in [0, 1]$ is an interpolation parameter that blends the nominal CV dynamics with the learned correction. Specifically, $\lambda = 0$ uses only the nominal model while $\lambda = 1$ fully adopts the learned transition.

Depending on the availability of training data, the structure of ΔF_k^{ML} can be further constrained. Under limited data, learning is restricted to diagonal velocity scaling,

$$\Delta F_k^{ML} = \text{diag}(\delta_{k,x}, \delta_{k,y}, \delta_{k,z}), \quad (28)$$

where δ_k 's are learned scalar correction factors applied to each velocity component. With sufficient data, however, full velocity coupling is permitted (i.e. entire $\mathbb{R}^{3 \times 3}$ matrix).

While Q_k in the CV model admits a closed-form parametric structure derived from a continuous-time white Gaussian acceleration disturbance model [36], such structure may no longer hold under model mismatch and unmodeled disturbances. Hence, to allow greater flexibility, the learned covariance Q_k^{ML} is parameterized using the Cholesky LDL composition described in Section III-E.

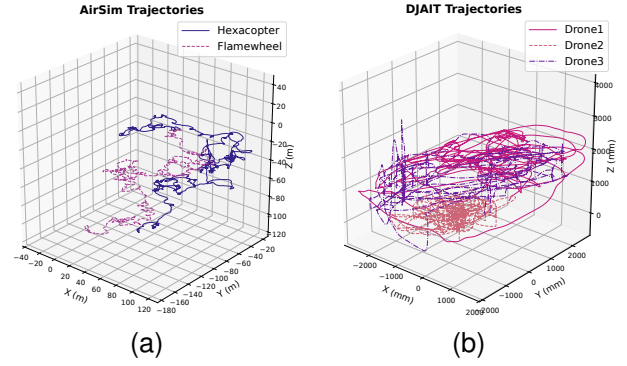


Fig. 3. Example trajectories used in the experiment. (a) Synthetic trajectories generated in AirSim for two quadrotor platforms. (b) Real-world trajectories from the DPJAIT dataset for three quadrotor platforms.

B. Datasets

We evaluate HAKF on both real-world flight data and high-fidelity simulation data. Example trajectories from both datasets are shown in Fig. 3. For real-world evaluation, we use the multimodal dataset indoor 3D drone tracking (DPJAIT) dataset [8], which provides ground-truth 3D trajectories from three different quadrotor platforms. For each recorded flight sequence, multiple reflective markers are attached to the drone body. In this work, the drone position is defined as the centroid of the tracked markers. The sampling interval is given by $dt = 1/f_s$, where $f_s = 100$ Hz [8].

In addition to real-world data, synthetic flight trajectories are generated using the Microsoft AirSim simulator for two quadrotor platforms, Flamewheel and Hexacopter. This allows evaluation in a large-data regime, allowing comparison with purely machine learning methods that typically rely on abundant training data. To sufficiently excite the system dynamics for model classification, trajectories are generated by continuously rotating the commanded horizontal velocity vector while modulating the yaw rate. This results in circular and gradually tightening horizontal motion patterns with oscillatory heading dynamics. These trajectories introduce non-constant velocity behavior, allowing the resulting motion patterns to exhibit distinct velocity dynamics versus simple waypoint-based motion.

To construct position-only observations for filtering, noisy measurements are generated by injecting zero-mean white Gaussian noise into the ground-truth position trajectories,

$$z_k = p_k + v_k, \quad v_k \sim \mathcal{N}(0, R_k), \quad (29)$$

where R_k is assumed to be known. Although the filter operates on position measurements, velocity ground truth is also required to evaluate full-state estimation accuracy performance, as described in Section IV-C1. Accordingly, velocity is computed from the position trajectory using finite differences.

C. Evaluation Metrics and Baseline Methods Comparison

The performance of the proposed HAKF is evaluated in terms of both state estimation and model classification.

1. Estimation

Estimation accuracy is measured using the root mean square error (RMSE) between the estimated state $\hat{x}_k$ and the ground-truth state x_k ,

$$\text{RMSE} = \sqrt{\frac{1}{T} \sum_{k=1}^T \|\hat{x}_k - x_k\|_2^2}. \quad (30)$$

To evaluate filter consistency, we compute the NEES and NIS statistics defined in (10)–(11). Consistency is assessed using a χ^2 hypothesis test with significance level 0.05. Specifically, we compute the proportion of NEES and NIS values that fall within the corresponding confidence interval and compare it with the ideal target coverage of 95%.

We compare the proposed HAKF with two baseline approaches. The first baseline is a classical CV KF with fixed transition matrix F and manually tuned process noise covariance Q , representing the standard model-based filtering approach commonly used for drone tracking. The second baseline is a purely learning-based model implemented using a recurrent neural network (RNN) [37] that directly estimates the system state from a window of recent measurements and is trained using ground-truth state supervision using (30). The RNN processes a fixed-length window of measurement increments and outputs the corresponding state estimate. Unlike KF methods, the RNN baseline produces only point estimates and does not provide a state covariance, and therefore statistical consistency metrics are not evaluated for this model.

2. Classification

Model classification performance is evaluated using trajectory-level accuracy. For each test trajectory the predicted label is produced once after processing the full sequence and is compared to the ground-truth label. The test set is approximately balanced across classes.

We compare the HAKF classification method based on GBI framework with three baseline approaches. The first baseline is a Random Forest [38] classifier that predicts the drone model directly from features extract from the measurement sequence. These features include summary statistics of the observed motion such as velocity and acceleration estimates derived from the measurements.

The second baseline augments the Random Forest with filtering-based features. In this approach, a classical CV KF is first applied to the measurement sequence and statistical quantities derived from the estimated filter states are used as input features for the Random Forest. This baseline evaluates whether incorporating filtering-based features alone can improve classification performance.

The third baseline is a purely learning-based model implemented using a RNN that processes the measurement sequence and outputs the predicted drone model. The network is trained using a cross-entropy loss function with the Adam optimizer.

D. Hyperparameter Tuning

To ensure fair comparison across methods, hyperparameters for all learning-based models were selected using grid search on validation datasets. For the RNN and the MLP component of HAKF, we varied the hidden dimension and number of layers. For the Random Forest classifiers, the number of trees and maximum tree depth were tuned.

V. Results

A. Estimation Performance

Table I reports the average state estimation accuracy and statistical consistency on test trajectories from both the DPJAIT real-world dataset and the AirSim simulation dataset, averaged over 20 Monte Carlo runs. We compare the proposed HAKF against the baseline methods described in Section IV-C1.

Starting with the DPJAIT dataset, the classical KF exhibits poor statistical consistency. In particular, the NEES value (60.12%) is significantly below the ideal target of 95%, indicating that fixed noise covariances cannot adequately account for time-varying model mismatch present in real flight dynamics. Although the NIS value is closer to the target, this is largely due to the assumption that R_k is known.

In contrast, the proposed HAKF substantially improves both estimation accuracy and statistical consistency. All HAKF variants achieve significantly lower RMSE compared to the baseline KF. Among the tested configurations, learning both the transition correction and process noise covariance with the NLL loss $(\Delta F, Q)^*$ achieves the best overall performance, reducing RMSE to 206.32 while also achieving the lowest NLL. Importantly, this configuration achieves estimation accuracy comparable to the model trained using RMSE loss, despite not requiring ground-truth state supervision during training.

To further evaluate performance when large training datasets are available, we consider the AirSim dataset containing approximately 10^5 training samples per drone. In this setting, the purely learning-based RNN model improves substantially compared to its performance on the DPJAIT dataset, achieving an RMSE of 88.39. This confirms that deep learning methods can outperform the classical KF baseline when sufficient training data are available. However, the proposed HAKF still achieves the best overall performance.

The same trend observed in the DPJAIT dataset persists, where the configuration learning both $(\Delta F, Q)^*$ provides strong statistical consistency (NEES 89.88%, NIS 94.03%) while maintaining RMSE performance compara-

TABLE I

Average tracking accuracy and statistical consistency on unseen drone trajectories from the DPJAIT and AirSim datasets, averaged over Monte Carlo runs. Training sample counts correspond to the number of points used to train the model.

Method	DPJAIT (Train: ≈ 5500 samples per drone)				AirSim (Train: ≈ 100000 samples per drone)			
	RMSE Loss $\downarrow$	NLL Loss $\downarrow$	NEES (%)	NIS (%)	RMSE Loss $\downarrow$	NLL Loss $\downarrow$	NEES (%)	NIS (%)
KF (Fixed F, Q) [‡]	311.44	12.71	60.12	92.67	160.54	14.24	66.12	91.92
HAKF _Q (Q only)*	210.59	12.07	83.22	92.27	95.03	9.73	81.68	93.48
HAKF ($\Delta F, Q$)*	206.39	11.98	86.32	93.56	83.07	9.59	89.88	94.03
HAKF ($\Delta F, Q$) [†]	206.32	12.05	83.68	92.36	82.69	9.64	85.92	93.86
RNN (Pure ML) [†]	327.92	–	–	–	88.39	–	–	–
Target	–	–	95	95	–	–	95	95

* Trained with NLL loss defined in (18). [†] Trained with RMSE loss defined in (30). [‡] KF F is fixed while static Q is manually tuned.

ble to the variant trained using RMSE loss. Furthermore, unlike RNN models which only provide point estimates, the HAKF produces full probabilistic state estimates, enabling uncertainty quantification and statistical consistency evaluation.

B. Classification Performance

Fig. 4 compares model classification accuracy across both simulated and real-world datasets. We compare HAKF’s GBI classification detailed in Section III-D with the baseline methods described in Section IV-C2.

We first analyze the AirSim dataset, where the number of training samples is systematically varied to study performance under different data regimes. In the low-data regime (fewer than 10^4 training samples), HAKF consistently achieves higher classification accuracy than the purely learning-based RNN model as well as the Random Forest baselines. This validates the advantage of incorporating model-based structure together with probabilistic inference through GBI when training data are limited in the classification setting.

As the amount of training data increases, the performance of HAKF plateaus and converges with that of the Random Forest methods. However, the RNN model improves substantially and eventually surpasses the other approaches in the high-data regime. This behavior is expected since deep neural networks can learn highly expressive nonlinear dynamics directly from data when sufficient training samples are available. In contrast, HAKF and Random Forest classifiers rely on structured features derived from filtering and measurement statistics, which introduce useful inductive bias in low-data settings but can limit representational capacity in the large-data regime. Despite this, the proposed HAKF maintains strong performance in the high-data regime, achieving approximately 84% classification accuracy.

We next evaluate classification performance on the DPJAIT real-world dataset using approximately 5500 training samples per drone. The results show that HAKF achieves the highest classification accuracy among all tested methods. In contrast, the RNN model exhibits the lowest performance on this dataset, indicating that purely

data-driven models struggle to generalize when training data are limited.

VI. Conclusion and Future Work

This article introduced a self-supervised HAKF framework that integrates model-based filtering with data-driven learning for both state estimation and classification. The proposed method augments a classical KF with a neural network that learns corrections to the system dynamics and process noise covariance using only measurement data. Within this framework, model classification is performed using GBI based on the innovation likelihood while maintaining the traditional filtering structure.

Experimental results on both real-world (DPJAIT) and simulated (AirSim) datasets demonstrate that the proposed framework improves estimation accuracy while achieving substantially better statistical consistency compared to classical KF. For classification, HAKF shows strong performance in low-data regimes where purely learning-based methods struggle while still remaining effective when larger training datasets are available. These results highlight the advantage of combining traditional model-based structure with data-driven adaptation across both low-data and high-data scenarios.

Despite these advantages, the proposed framework has several limitations. The adaptive dynamics correction is structured and therefore cannot represent arbitrarily complex nonlinear dynamics. Moreover, the current formulation still relies on standard KF assumptions such as white Gaussian noise. As a result, this method may fail when the incorporated inductive bias is fundamentally incorrect, and purely learning-based models may outperform when very large training datasets are available. Future work will explore extending the proposed framework to more expressive dynamical models and nonlinear filtering formulations in order to address the limitations and assumptions embedded in classical KF.

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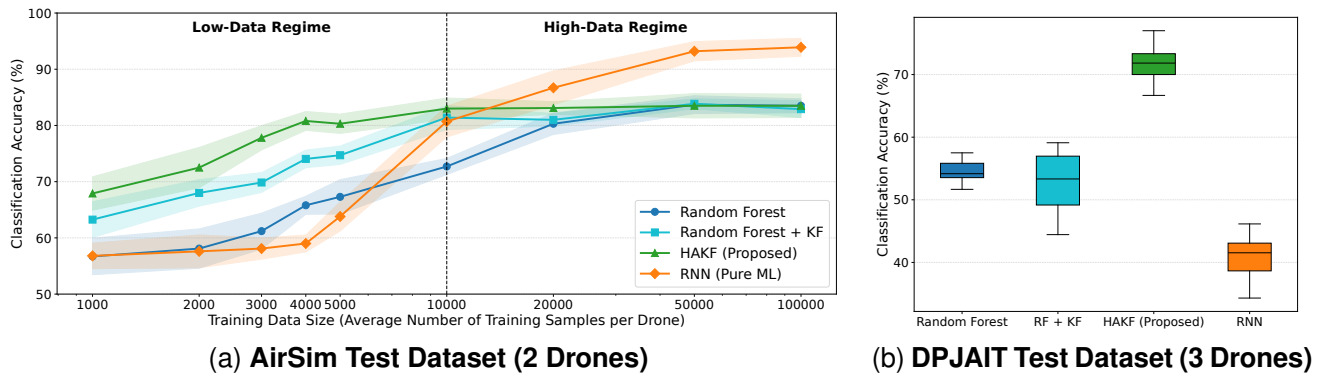


Fig. 4. Model classification accuracy across datasets. (a) AirSim simulated dataset showing performance as a function of average training samples per drone. (b) DPJAIT real-world dataset using approximately 5500 training samples per drone.

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